# Notes: Sarsons, Gërxhani, Reuben, and Schram (JPE, 2021)

Gender Differences in Recognition for Group Work

## Contents

# 1 Big picture

- **Question.** Does gender affect how credit is assigned for group work when individual contributions are not directly observed? The paper studies whether women receive less recognition for jointly produced output, and whether this can help explain promotion gaps.
- **Main setting.** The core observational setting is academic economics. The authors construct CV-based data for economists who came up for tenure between 1985 and 2014 at top 35 US PhD-granting economics departments.
- **Central empirical contrast.** Solo-authored papers provide relatively clear signals of individual ability. Coauthored papers provide noisier signals because evaluators must infer how much each author contributed.
- **Main observational result.** Men receive similar tenure returns from solo-authored and coauthored papers. Women receive strong returns from solo-authored papers, but weaker returns from coauthored papers, especially papers coauthored with men.
- **Magnitude.** In the main paper-composition specification, an additional coauthored paper is associated with a 7.4 percentage point increase in tenure probability for men but only a 4.7 percentage point increase for women.
- **Coauthor-gender result.** The lower return to coauthorship for women is driven mainly by papers written with male coauthors. Papers coauthored with women do not show the same penalty.
- **Decomposition result.** A counterfactual exercise suggests that unequal credit attribution accounts for about 39% of the unconditional gender tenure gap and about 65% of the conditional gap after controlling for paper quality, citations, institution, field, and related observables.
- **Experimental result.** Two experiments show that when evaluators observe a joint output but not individual contributions, they attribute credit differently by gender. The pattern depends on stereotypes about the task and on the evaluator's gender and beliefs.
- **Economic takeaway.** Group work can magnify inequality when evaluators must infer contributions. The issue is not simply that women collaborate more or produce less; rather, noisy signals allow gendered beliefs to enter promotion and hiring decisions.

# 2 Incremental contribution of the paper

- **From promotion gaps to credit attribution.** Prior work documents gender gaps in promotion and tenure. This paper identifies a specific mechanism: women may receive less credit for team output when individual contributions are hard to observe.
- **A clean empirical contrast within academia.** The paper uses the distinction between solo-authored and coauthored papers as a natural contrast between clearer and noisier signals of individual contribution.
- **A coauthor-gender contribution.** The authors do not only compare solo versus coauthored papers. They also split coauthored papers by the gender composition of coauthors, showing that the penalty for women is concentrated in joint work with men.

- **A decomposition contribution.** The paper quantifies how much of the tenure gap could be explained by different credit assignment for solo and coauthored papers.
- **A mechanism contribution.** The CV evidence is combined with tests of alternative stories: selection into coauthorship, coauthor seniority, presentation behavior, ability-based sorting, timing of collaboration, and post-tenure productivity.
- **A multi-method contribution.** The paper pairs observational evidence with two experiments. The experiments shut down selection and effort channels by design and show that credit-attribution bias can arise even when individual contributions are exogenously hidden.
- **A beliefs contribution.** The experiments show that credit attribution is linked to beliefs about which gender is better at a task. This connects the tenure evidence to statistical discrimination and stereotype-based inference.
- **A broader labor-market contribution.** The paper matters beyond academia because many organizations increasingly rely on teams. If team output is evaluated without transparent contribution measurement, biased credit attribution can affect promotions, hiring, and occupational segregation.

# 3 Data and measurement

## 3.1 CV data and tenure sample

- The sample contains economists who came up for tenure between 1985 and 2014 at one of the top 35 US PhD-granting economics departments.
- The unit of observation is an individual economist's tenure case.
- The authors collect CV information on PhD institution and year, employment history, publication history, fields, and coauthors.
- The main sample contains 613 economists: 476 men and 137 women.
- The setting is useful because tenure decisions at research universities place large weight on publications, and economics traditionally uses alphabetical author ordering rather than explicit author-contribution ordering.

**Interpretation:** The data allow the authors to compare researchers with similar observed publication records but different solo/coauthored paper composition. The design is strongest when tenure committees evaluate output in a similar institutional setting.

## 3.2 Tenure construction

- Tenure is coded using institutional tenure timelines, typically around years 6–8 after initial appointment.
- An individual is treated as denied tenure if they move to a university ranked more than five positions below the initial institution during the tenure window, or if they move from academia to industry during the tenure window.
- Moves to comparable or better institutions during the tenure window are not treated as denials, because such moves may reflect outside opportunities rather than failure.

- If someone moves before the tenure window, the authors use the second institution to determine tenure, without restarting the tenure clock in the main specification.

**Interpretation:** The tenure variable is necessarily reconstructed from careers rather than directly observed committee votes. The robustness analysis addresses concerns that the classification or missing post-denial CVs drive the results.

### 3.3 Publication and quality measures

- Publications are counted up to and including the tenure year in the main analysis.
- Papers are split into solo-authored and coauthored papers.
- Paper quality is measured primarily using the Kalaitzidakis, Mamuneas, and Stengos AER-equivalent journal ranking.
- Citations to pretenure papers are measured using Google Scholar citations scraped in 2015.
- The analysis also controls for total coauthors, years to tenure, field fixed effects, tenure-institution fixed effects, and tenure-year fixed effects.

**Interpretation:** The main comparison is not simply between more productive and less productive researchers. The authors attempt to compare men and women with similar numbers and quality of papers, citations, fields, institutions, and tenure timing.

### 3.4 Experimental data

- **Experiment I** uses mTurk workers and predictors. Workers complete quizzes individually; predictors then estimate future quiz performance after observing either individual scores or combined male-female scores.
- Experiment I varies three dimensions: individual versus joint score information, no gender information versus gender score-distribution information, and male-stereotyped math quizzes versus female-stereotyped grammar quizzes.
- **Experiment II** uses actual HR personnel as recruiters. Recruiters choose candidates from short resumes and are paid based on the chosen candidate's task score.
- Experiment II varies individual versus joint score information and uses search and vocabulary tasks. It also elicits recruiters' beliefs about gender differences in task performance.

**Interpretation:** The experiments complement the CV data because they remove two major confounds: real differences in contribution and endogenous selection into teams.

## 4 Models and empirical specifications

### 4.1 Tenure gap and total publications

The first specification relates tenure to total pretenure publications:

$$T_{ifst} = \beta_1 TotPapers_i + \beta_2 TotPapers_i^2 + \gamma' Z_i + \nu_f + \nu_s + \nu_t + \varepsilon_{ifst}.$$

- $T_{ifst}$ is an indicator for receiving tenure.
- $TotPapers_i$ is the number of solo and coauthored papers at tenure.

- $Z_i$ includes paper quality, log citations, years to tenure, and total coauthors.
- Field, tenure-institution, and tenure-year fixed effects absorb systematic differences in standards.

**Interpretation:** This specification asks whether men and women with similar overall publication portfolios have similar tenure probabilities. The answer is no: women remain less likely to receive tenure conditional on total papers and controls.

### 4.2 Paper-composition specification

The central specification separates solo-authored and coauthored papers:

$$T_{ifst} = \beta_1 S_i + \beta_2(Fem_i \times S_i) + \beta_3 CA_i + \beta_4(Fem_i \times CA_i) + \delta Fem_i + \gamma' Z_i + \nu_f + \nu_s + \nu_t + \varepsilon_{ifst}.$$

- $S_i$ is the number of solo-authored papers.
- $CA_i$ is the number of coauthored papers.
- $Fem_i \times S_i$ asks whether solo papers matter differently for women.
- $Fem_i \times CA_i$ asks whether coauthored papers matter differently for women.

**Interpretation:** Solo papers help women close the tenure gap. Coauthored papers help women less than men. This is the core evidence that noisy team signals are treated differently by gender.

### 4.3 Coauthor-gender specification

The authors then split coauthored papers by the gender composition of coauthors:

$$\begin{aligned} T_{ifst} = \beta_1 S_i + \beta_2(Fem_i \times S_i) + \beta_3 CA_i^{male} + \beta_4(Fem_i \times CA_i^{male}) \\ + \beta_5 CA_i^{mix} + \beta_6(Fem_i \times CA_i^{mix}) + \beta_7 CA_i^{female} + \beta_8(Fem_i \times CA_i^{female}) \\ + \beta_9 Fem_i + \gamma' Z_i + \nu_f + \nu_s + \nu_t + \varepsilon_{ifst}. \end{aligned}$$

- $CA_i^{male}$ counts papers with only male coauthors.
- $CA_i^{female}$ counts papers with only female coauthors.
- $CA_i^{mix}$ counts papers with both male and female coauthors.

**Interpretation:** The coauthor-gender split is crucial. If all coauthorship reduced women's tenure chances equally, the result could be about collaboration preferences or work style. Instead, the penalty is concentrated in work with men.

### 4.4 Counterfactual decomposition

- The authors first estimate tenure probabilities allowing for a female intercept but not different returns to solo and coauthored papers.
- They then estimate a model allowing solo and coauthored papers to matter differently for women.
- They predict what women's tenure rates would have been if their papers were treated like men's papers.

- The exercise suggests that different credit attribution closes the tenure gap by about 8.5 percentage points.

**Interpretation:** The decomposition is not a randomized causal decomposition. It is an accounting exercise showing that the estimated difference in returns to coauthored work is large enough to explain a meaningful share of the gender tenure gap.

### 4.5 Experiment I prediction model

Experiment I estimates predicted quiz-2 scores as a function of the worker's gender and information treatment:

$$Q2_{ij} = \beta_1 Fem_i + \beta_2 D_j + \beta_3 (Fem_i \times D_j) + \beta_4 Q1_i + \varepsilon_{ij}.$$

- $Q2_{ij}$ is predictor $j$'s estimate of worker $i$'s second quiz score.
- $Fem_i$ indicates that the worker is female.
- $D_j$ indicates that the predictor receives gender-performance information.
- The model is estimated separately for math versus grammar and individual versus joint score conditions.

**Interpretation:** When individual scores are visible, gender does not significantly affect predicted math performance. When only joint scores are visible, women are predicted to do worse on math. This mirrors the solo/coauthored contrast in the tenure data.

### 4.6 Experiment II recruiter-choice model

Experiment II uses a conditional logit model for recruiter choices:

$$u_{jik} = \beta_1 Fem_{ik} + \beta_2 (Fem_{ik} \times Belief_j) + \beta_3 Score_{ik} + \gamma' Z_{ik} + \nu_{jk} + \varepsilon_{jik}.$$

- Recruiter $j$ chooses one candidate $i$ from candidate set $k$.
- $Fem_{ik}$ indicates a female candidate.
- $Belief_j$ measures the recruiter's belief about male versus female performance.
- $Score_{ik}$ is the observed joint score in the joint-score condition.
- Set-by-recruiter fixed effects compare candidates within the same choice set.

**Interpretation:** The experiment shows that credit attribution in hiring-like settings is mediated by beliefs. Recruiters who believe men are better are less likely to choose female candidates from joint outputs, and vice versa.

## 5 Identification and empirical design

### 5.1 What the observational design can identify

- The CV analysis identifies conditional associations between paper composition and tenure outcomes.

- It can show that women and men with similar observed publication portfolios receive different tenure returns to coauthored work.
- It cannot by itself prove that tenure committees directly discriminate, because actual individual contributions to coauthored papers are unobserved.
- The paper therefore builds a cumulative case using robustness checks, mechanism tests, and experiments.

**Interpretation:** The correct reading is not that the CV evidence alone fully proves bias. The correct reading is that the observed pattern is difficult to reconcile with many non-bias explanations and is supported by experimental evidence.

## 5.2 Threats and alternative explanations

- **Attrition.** If denied-tenure economists disappear from the sample, the estimates could be biased. The authors address this with historical faculty lists.
- **Paper quality.** If women's coauthored papers are lower quality, the coauthor penalty could reflect productivity. The authors control for journal rank and citations and use alternative ranking measures.
- **Coauthor seniority.** If women disproportionately coauthor with senior men who receive more credit, this could explain the result. The data do not support this story.
- **Sorting by ability.** If lower-ability women select into coauthoring with high-ability men, the penalty might be rational. The paper finds little evidence of such sorting.
- **Credit claiming.** If women present joint work less often, they may receive less recognition. Survey evidence does not show that women are less likely than coauthors to present joint work.
- **Taste-based discrimination.** If committees simply dislike tenuring women, solo-authored women should also be penalized. But women who solo author enough papers have tenure rates comparable to men's.

**Interpretation:** No single observational test is decisive, but the pattern across tests points toward biased inference about women's contributions to joint work rather than lower contribution or rational sorting.

## 5.3 Experimental identification

- The experiments randomly vary whether evaluators see individual output or joint output.
- Workers complete tasks individually and are individually incentivized, so hidden joint scores cannot be caused by free riding.
- Pairing is random, so hidden joint scores cannot be caused by endogenous team formation.
- The experiments therefore isolate the evaluator's inference problem: how much credit should each person receive when only combined output is visible?

**Interpretation:** The experiments are not identical to tenure decisions, but they directly test the mechanism that the tenure data suggest: when contribution signals are noisy, gendered beliefs can shape credit assignment.

### 5.4 External validity

- The academic tenure setting is high-stakes and specialized, while the experiments are simplified and lower-stakes.
- Economics may be perceived as a male-stereotyped field, so the experimental math-task result may be especially relevant.
- The HR experiment broadens the setting by using professional recruiters rather than only online predictors.
- The long-run implications are broader because team production is common in academia, technology, consulting, finance, and many other organizations.

**Interpretation:** The paper's external validity is strongest for environments with team output, imperfect observability of individual contributions, and gendered beliefs about task ability.

## 6 Tables and figures

### 6.1 Table 1: Summary statistics

**Read:**

- The full sample contains 613 economists, of whom 476 are men and 137 are women.
- The average tenure rate is 68.7%, but there is a large gender gap: 73.5% for men versus 51.8% for women.
- Men and women have similar numbers of total papers, solo-authored papers, and coauthored papers at tenure.
- The main statistically significant differences are not in paper counts but in tenure outcomes and some coauthoring patterns.
- Women have more female coauthors than men do, consistent with gendered collaboration patterns.

**Economic message:** Table 1 establishes the puzzle. Women are much less likely to receive tenure even though their observed pretenure publication counts are broadly similar to men's. The rest of the paper asks whether the type of output signal - solo versus coauthored - explains part of this gap.

### 6.2 Figure 1 and Figure 2: Baseline tenure gap and paper composition

**Read:**

- Figure 1 plots tenure against total number of publications, residualizing for controls. Both men and women have higher tenure probabilities with more papers.
- In Figure 1, women remain below men at comparable total publication levels, suggesting that total output does not eliminate the gender tenure gap.
- Figure 2 replaces total publications with the fraction of papers that are solo-authored.
- For men, the fraction solo-authored is almost unrelated to tenure probability.
- For women, the relationship is strongly positive: women with more solo-authored portfolios have substantially higher tenure probabilities.

TABLE 1
SUMMARY STATISTICS

| | Full | Male | Female | *p*-Value |
|---|---|---|---|---|
| | A. Sample Characteristics | | | |
| Tenure | .687 | .735 | .518 | .001 |
| | (.464) | (.442) | (.502) | |
| Years to tenure | 6.770 | 6.637 | 7.234 | .001 |
| | (1.645) | (1.579) | (1.787) | |
| Total papers | 8.421 | 8.517 | 8.088 | .259 |
| | (3.916) | (4.078) | (3.282) | |
| Solo authored | 3.055 | 3.048 | 3.080 | .890 |
| | (2.371) | (2.406) | (2.253) | |
| Coauthored | 5.364 | 5.464 | 5.015 | .194 |
| | (3.573) | (3.697) | (3.089) | |
| | B. Quality of Papers | | | |
| Top five solo | .651 | .643 | .679 | .704 |
| | (.976) | (.988) | (.939) | |
| Top five coauthored | 1.266 | 1.275 | 1.234 | .764 |
| | (1.426) | (1.430) | (1.416) | |
| AER-equivalent ranking | .331 | .338 | .309 | .068 |
| | (.161) | (.161) | (.160) | |
| AER-equivalent: | | | | |
| Solo publications | .317 | .314 | .326 | .599 |
| | (.238) | (.232) | (.256) | |
| Coauthored publications | .320 | .328 | .290 | .058 |
| | (.205) | (.209) | (.189) | |
| | C. Coauthoring Patterns | | | |
| Number of unique coauthors | 4.551 | 4.562 | 4.512 | .860 |
| | (2.807) | (2.792) | (2.867) | |
| Full professor | .488 | .499 | .450 | .104 |
| | (.306) | (.315) | (.270) | |
| Associate professor | .163 | .166 | .152 | .531 |
| | (.227) | (.235) | (.195) | |
| Assistant professor | .279 | .270 | .311 | .140 |
| | (.282) | (.279) | (.290) | |
| Graduate student | .0159 | .0138 | .0233 | .158 |
| | (.0681) | (.0565) | (.0984) | |
| Number of female coauthors | .1342 | .0927 | .279 | .001 |
| | (.229) | (.180) | (.310) | |
| Observations | 613 | 476 | 137 | |

NOTE.—This table displays the average tenure rate, pretenure productivity, and pretenure authorship patterns of men and women who went up for tenure at one of the top 35 US economics departments between 1985 and 2014. The top 35 institutions are determined according to the RePEc/IDEAS economics department rankings. In panel A, "Tenure" is an indicator that equals one if an individual was promoted to associate or full professor 6–8 years after his or her initial appointment. "Years to tenure" gives the number of years between an individual's PhD graduation year and the year he or she went up for tenure. All paper counts are measured as the number of papers an individual had published at the time of tenure. "Top five solo" and "Top five coauthored" give the number of publications an individual had published in one of the top five economics journals: *AER*, *Quarterly Journal of Economics*, *Econometrica*, *Journal of Political Economy*, and *Review of Economic Studies*. "AER equivalent" is a measure that converts an individual's publications into the number of AER-equivalent publications they correspond to. For more details on this variable, see sec. II. "Number of unique coauthors" gives the number of different coauthors an individual had published with by the time he or she went up for tenure. Coauthor positions (full, associate, assistant, and graduate student) are the positions an individual's coauthors had at the time that individual went up for tenure.

Table 1: Summary Statistics.

**Economic message:** The visual evidence captures the paper's central insight. Total output helps both genders, but paper composition matters much more for women. Clearer individual signals close the gap; noisier team signals do not.

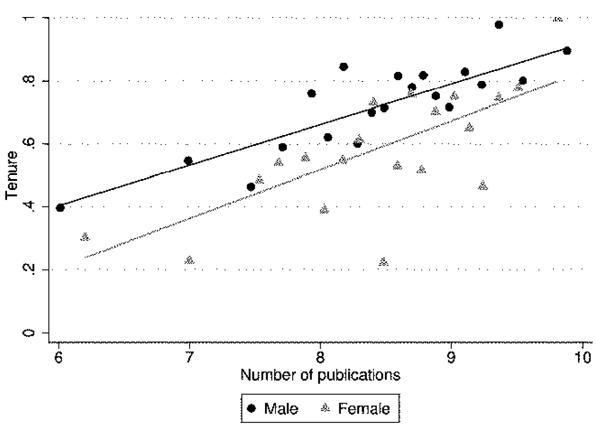


Fig. 1.—Total papers and tenure. This binned scatterplot shows the correlation between the total number of publications an individual has at the time they go up for tenure and the probability of receiving tenure. The *y*-axis variable, tenure, is a binary variable that equals one if an individual received tenure at their initial institution of employment. For more details on how the tenure variable is constructed, see section II. To construct the plot, tenure is first residualized with respect to the following controls: number of years it took to go up for tenure, average journal rank of pretenure publications, log citations, total coauthors, and tenure school group, tenure year group, and field fixed effects. The *x*-axis variable, number of publications, is then divided into 20 equal-sized groups. Within each of these groups, we plot the mean of the *y*-axis variable (tenure) residuals against the mean of the *x*-axis variable (also within each bin). We then add back the unconditional mean of tenure to help with the interpretation of the line of best fit. The lines of best fit are estimated using the full sample ($N = 613$) and have slopes of $\beta = 0.129$ (SE = 0.016) for men and $\beta = 0.155$ (SE = 0.040) for women. A color version of this figure is available online.

(a) Figure 1: Total papers and tenure.

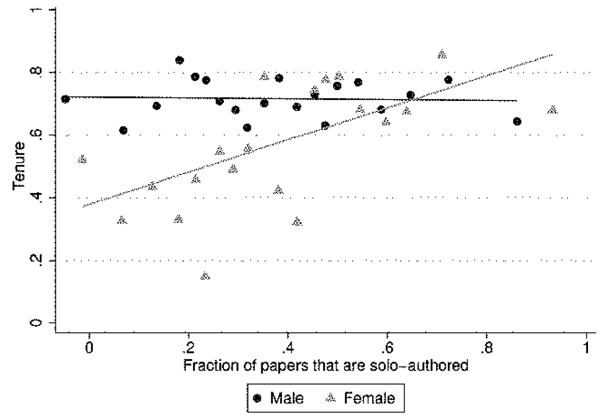


Fig. 2.—Relationship between paper composition and tenure. This figure is a binned scatterplot of the correlation between tenure and the fraction of an individual's papers that are solo authored, split by gender. The *y*-axis variable is a binary variable indicating whether an individual received tenure. To construct the plot, tenure is first residualized with respect to the following controls: total number of papers an individual published by the time of tenure, number of years it took to go up for tenure, average journal rank of pretenure publications, log citations, total coauthors, and tenure school group, tenure year group, and field fixed effects. The *x*-axis variable, fraction of papers that are solo authored, is then divided into 20 equal-sized groups. Within each of these groups, we plot the mean of the *y*-axis variable (tenure) residuals against the mean of the *x*-axis variable (also within each bin). We then add back the unconditional mean of tenure to help with the interpretation of the line of best fit. The line of best fit using OLS is shown separately for men and women. The lines of best fit are estimated using the full sample ($N = 613$) and have slopes of $\beta = 0.514$ (SE = 0.154) for women and $\beta = -0.013$ (SE = 0.069) for men. A color version of this figure is available online.

(b) Figure 2: Paper composition and tenure.

## 6.3 Table 2 and Table 3: Main tenure regressions and coauthor gender

**Read:**

- Table 2 first shows that the marginal relationship between total papers and tenure is not significantly different for women and men.
- Once papers are split by authorship, solo-authored papers help women more than men because women start from a lower baseline.
- Coauthored papers help both groups, but help men more. In the preferred specification, a coauthored paper is worth about 7.4 percentage points for men and about 4.7 percentage points for women.
- Table 3 shows that the coauthor penalty for women is concentrated in papers written with male coauthors.
- A paper with only male coauthors is associated with a 7.8 percentage point tenure increase for men but only about 1.8 percentage points for women.

**Economic message:** Table 2 is the core regression evidence. Table 3 sharpens the mechanism: the problem is not coauthorship in general but credit attribution in mixed-gender or male-coauthored work.

TABLE 2
Relationship between Papers and Tenure

| | Dependent Variable: Tenure | | | | |
|---|---|---|---|---|---|
| | Full (1) | Full (2) | Full (3) | Female (4) | Male (5) |
| Total papers | .133*** | | | | |
| | (.016) | | | | |
| Total papers$^2$ | −.004*** | | | | |
| | (.001) | | | | |
| Female × papers | −.005 | | | | |
| | (.009) | | | | |
| Solo authored | | .096*** | .095*** | .201*** | .088*** |
| | | (.016) | (.016) | (.050) | (.021) |
| Total solo$^2$ | | −.005*** | −.005*** | −.007 | −.004*** |
| | | (.001) | (.001) | (.005) | (.002) |
| Female × solo | | .048*** | .060*** | | |
| | | (.014) | (.012) | | |
| Coauthored | | .086*** | .074*** | −.021 | .084*** |
| | | (.017) | (.013) | (.055) | (.010) |
| Coauthored$^2$ | | −.003*** | −.002*** | −.000 | −.003*** |
| | | (.001) | (.001) | (.004) | (.001) |
| Female × coauthored | | −.028** | −.027** | | |
| | | (.014) | (.012) | | |
| Total coauthors | −.004 | .001 | .004 | .026 | −.001 |
| | (.003) | (.005) | (.004) | (.018) | (.003) |
| Log citations | .052*** | .031** | .057*** | .077 | .053*** |
| | (.014) | (.014) | (.014) | (.066) | (.013) |
| AER-equivalent ranking | .542*** | | | | |
| | (.113) | | | | |
| AER-equivalent solo | | .154* | .338*** | .295 | .424*** |
| | | (.090) | (.071) | (.275) | (.060) |
| AER-equivalent coauthored | | .210** | .352*** | .577** | .300*** |
| | | (.085) | (.064) | (.256) | (.078) |
| Female | −.133 | −.183* | −.216** | | |
| | (.084) | (.102) | (.090) | | |
| Tenure institution fixed effects | Yes | No | Yes | Yes | Yes |
| Tenure year fixed effects | Yes | No | Yes | Yes | Yes |
| Field fixed effects | Yes | No | Yes | Yes | Yes |
| Observations | 612 | 608 | 608 | 137 | 471 |
| $R^2$ | .411 | .289 | .428 | .518 | .425 |

Note.—This table shows the relationship between publications and tenure. The dependent variable, "Tenure," is binary and indicates whether an individual received tenure 6–8 years after being hired at the initial tenure institution. "Total papers" gives the number of papers an individual published by the time he or she went up for tenure. "Solo authored" and "Coauthored" give the number of solo- or coauthored papers he or she had published at the time of tenure. "AER-equivalent ranking," "AER-equivalent solo," and "AER-equivalent coauthored" are journal quality measures described in sec. II. "Total coauthors" gives the number of coauthors an individual had on the papers he or she had published by the time of tenure. "Tenure length" gives the number of years it took the individual to go up for tenure. Citations are from Google Scholar and were measured in 2015. The equations are estimated using a linear probability model. Bootstrapped standard errors are clustered by tenure institution and reported in parentheses.

* $p < .10$.
** $p < .05$.
*** $p < .01$.

(a) Table 2: Relationship between papers and tenure.

TABLE 3
Coauthor Gender

| | (1) | × Female (2) |
|---|---|---|
| Solo authored | .092*** | .052*** |
| | (.015) | (.012) |
| Papers with only female coauthors | .084** | .019 |
| | (.027) | (.024) |
| Papers with only male coauthors | .078*** | −.060*** |
| | (.013) | (.015) |
| Papers with male and female coauthors | .082*** | .031 |
| | (.024) | (.032) |
| Female | −.148 | |
| | (.093) | |
| Total coauthors | .001 | |
| | (.004) | |
| Log citations | .057*** | |
| | (.014) | |
| AER-equivalent coauthored | .366*** | |
| | (.062) | |
| AER-equivalent solo | .344*** | |
| | (.070) | |
| Tenure institution fixed effects | Yes | |
| Tenure year fixed effects | Yes | |
| Field fixed effects | Yes | |
| Observations | 606 | |

Note.—This table presents the results of one regression where the variables that are interacted with "Female" (a dummy indicating that the researcher is a woman) are displayed in the right-hand column. "Papers with only female coauthors" gives the number of publications an individual has in which all coauthors are female. Similarly, "Papers with only male coauthors" and "Papers with male and female coauthors" give the number of publications with only male coauthors and with a mix of male and female coauthors, respectively. Controls for tenure length, quadratics in the number of papers, and tenure institution group, tenure year group, and field fixed effects are also included. The equations are estimated using a linear probability model. Bootstrapped standard errors are reported in parentheses and are clustered by tenure institution.

** $p < .05$.
*** $p < .01$.

(b) Table 3: Coauthor gender.

## 6.4 Table 4: Robustness checks

**Read:**

- Table 4 tests whether the main result survives changes in sample and measurement.
- Column 1 uses only the faculty-list sample, where missing denied-tenure cases should be less problematic.
- Columns 2–4 use alternative journal-ranking controls: RePEc rankings, rankings that vary over time, and AER-equivalent bins.
- Columns 5–6 include papers published one or two years after tenure, allowing for papers that were likely in the pipeline during review.
- Column 7 uses total AER-equivalent publication counts rather than raw paper counts.
- Across specifications, the interaction between female and coauthored papers remains negative.

**Economic message:** The robustness table shows that the main pattern is not an artifact of one journal-quality measure, tenure-window definition, or sample construction. The coauthorship penalty for women is persistent.

TABLE 4
ROBUSTNESS CHECKS

| | | Journal Rankings | | | Publication Count | | |
|---|---|---|---|---|---|---|---|
| | Faculty List Sample (1) | RePEc (2) | Over Time (3) | AER Bins (4) | Tenure +1 (5) | Tenure +2 (6) | Total AERs (7) |
| Solo authored | .111*** | .088*** | .076*** | .074*** | .055*** | .043*** | .059*** |
| | (.014) | (.016) | (.017) | (.019) | (.012) | (.012) | (.017) |
| Female × solo | .051*** | .059*** | .055*** | .047*** | .047*** | .057*** | .102*** |
| | (.011) | (.012) | (.012) | (.012) | (.012) | (.012) | (.020) |
| Coauthored | .078*** | .072*** | .072*** | .060*** | .035*** | .032*** | .086*** |
| | (.018) | (.013) | (.014) | (.018) | (.009) | (.006) | (.014) |
| Female × coauthored | −.035** | −.029** | −.026** | −.029** | −.022** | −.013 | −.035* |
| | (.016) | (.011) | (.011) | (.012) | (.010) | (.011) | (.019) |
| Years to tenure | −.042*** | −.052*** | −.053*** | −.049*** | −.043*** | −.044*** | −.050*** |
| | (.011) | (.010) | (.010) | (.009) | (.009) | (.009) | (.009) |
| Total coauthors | −.001 | .007* | .005 | −.003 | .009** | .010*** | .003 |
| | (.004) | (.004) | (.005) | (.006) | (.004) | (.003) | (.004) |
| Log citations | .053*** | .062*** | .069*** | .071*** | .064*** | .062*** | .085*** |
| | (.014) | (.013) | (.015) | (.014) | (.015) | (.013) | (.019) |
| AER-equivalent coauthored | .357*** | .001* | .003*** | | .371*** | .341*** | |
| | (.049) | (.001) | (.001) | | (.070) | (.072) | |
| AER-equivalent solo | .429*** | .003*** | .004*** | | .301*** | .323*** | |
| | (.072) | (.001) | (.001) | | (.073) | (.074) | |
| Female | −.150* | −.203** | −.206** | −.183** | −.208*** | −.291*** | −.208*** |
| | (.087) | (.084) | (.085) | (.077) | (.078) | (.094) | (.063) |
| Observations | 362 | 612 | 612 | 612 | 608 | 595 | 612 |

NOTE.—The dependent variable in all columns is an indicator for receiving tenure. Column 1 restricts the sample to those schools we received a historical faculty list from. Column 2 uses RePEc journal rankings as the paper quality measure. The ranking used can be found at https://ideas.repec.org/top/top.journals.all.html. Column 3 uses historical journal rankings from RePEc to allow for rankings to change over time and to account for new journals entering. Column 4 controls for the number of papers within each of 10 AER "bins." For this analysis, the AER-equivalent measure is divided into deciles. For each individual, we then add up the number of solo- and coauthored papers within each decile and include the number of papers in each bin as controls. In cols. 5 and 6, we include papers that were published 1 and 2 years after an individual went up for tenure in the paper counts. In col. 7, we use the AER-equivalent measure of journal ranking to calculate the total number of AER equivalents (solo and coauthored) an individual had at the time of tenure. We use this measure in place of the solo- and coauthored paper counts (the main independent variables). All regressions control for a quadratic in the number of papers and tenure length, as well as tenure institution group, tenure year group, and field fixed effects. Bootstrapped standard errors are reported in parentheses and are clustered by tenure institution.

* $p < .10$.
** $p < .05$.
*** $p < .01$.

Table 4: Robustness Checks.

## 6.5 Table 5 and Table 6: Future productivity and survey evidence

**Read:**

- Table 5 uses post-tenure-decision productivity to ask whether women who coauthor and are denied tenure look less productive afterward.

- The estimates are noisy, but they do not show that denied women who coauthor are clearly lower productivity than denied men.
- Table 6 reports survey evidence on economists' beliefs about the tenure returns to different papers.
- Men and women report very similar beliefs about the return to a coauthored AER: 12.1% for men versus 12.2% for women.
- Women also do not report being less likely than their coauthors to present joint work.

**Economic message:** These tables weaken two alternative stories. The results are not obviously driven by women contributing less, and women do not appear to anticipate that coauthored work receives lower tenure credit.

TABLE 5
FUTURE PRODUCTIVITY

| | Outcome Variable: Post–Tenure Decision Solo AER Equivalents Poisson (1) | Outcome Variable: Log Citations OLS (2) |
|---|---|---|
| Fraction coauthored | −1.247* | .443 |
| | (.592) | (.383) |
| Female | −.136 | −.178 |
| | (.352) | (.418) |
| Female × fraction coauthored | −.018 | .732 |
| | (.604) | (.656) |
| Tenured | .062 | .328 |
| | (.390) | (.273) |
| Tenured × fraction coauthored | −.545 | .470 |
| | (.553) | (.463) |
| Female × tenured | .412 | .200 |
| | (.431) | (.519) |
| Female × tenured × fraction coauthored | −.391 | −.624 |
| | (.927) | (.780) |
| Top five coauthored | .016 | |
| | (.010) | |
| Total papers | | .075*** |
| | | (.016) |
| Observations | 602 | 612 |

NOTE.—Column 1 shows the results from estimating eq. (6) using a zero-inflated Poisson model, where the outcome variable is the number of solo-authored AER equivalents an individual published after the tenure decision (measured as of 2017). "Top five coauthored" gives the number of coauthored AER equivalents the individual published after tenure. Post–tenure decision institution is the institution the individual went to following the tenure decision. For people who received tenure, this is the same as the tenure institution. Column 2 shows the results from estimating the same equation using OLS, where "Log Citations" is the outcome variable. Citations are measured in 2015. Robust standard errors are reported in parentheses and are clustered at the tenure or post–tenure decision institution level.
* $p < .10$.
*** $p < .01$.

(a) Table 5: Future productivity.

TABLE 6
SURVEY RESULTS

| | Men (1) | Women (2) | $p$-Value (3) |
|---|---|---|---|
| | A. Beliefs about Returns to Papers | | |
| Coauthored AER | 12.1 | 12.2 | .939 |
| Coauthored AER, senior faculty | 9.1 | 8.8 | .528 |
| Coauthored AER, junior faculty | 13.3 | 13.4 | .796 |
| Solo top field | 8.0 | 8.2 | .669 |
| Coauthored top field | 6.3 | 6.8 | .223 |
| | B. Frequency of Presenting Papers | | |
| Times presented | 3.1 | 2.2 | .071 |
| Present more frequently than coauthor | .37 | .44 | .203 |
| Observations | 300 | 89 | |

NOTE.—This table presents the mean responses for men and women to two survey questions. For panel A, "Suppose a solo-authored AER increases your chance of receiving tenure by 15%. By how much do you think each of the following increases your change of receiving tenure?" For panel B, "How many times per year do you typically present your solo-authored papers? Are you more or less likely than your coauthors to present a joint paper?" "Present more frequently than coauthor" gives the fraction of respondents who reported that they are more likely than their coauthors to present a joint paper. The survey was conducted with a sample of academic economists currently working at a top 35 US economics department. Respondents were anonymous.

(b) Table 6: Survey results.

## 6.6 Figure 3 and Figure 4: Ability sorting and assortative matching

**Read:**

- Figure 3 uses the rank of the job-market-paper publication outlet as a proxy for individual ability.
- Panel A shows no meaningful relationship between this ability proxy and the fraction of papers that are coauthored.
- Panel B shows that women are more likely to coauthor with women, but high-ability women are not especially likely to avoid male coauthors.
- Figure 4 shows positive assortative matching by ability: higher-ability economists tend to have higher-ability coauthors.

- However, the paper shows that coauthor ability or prominence does not explain the tenure gap for coauthoring women.

**Economic message:** If low-ability women were the ones selecting into coauthorship with men, the tenure penalty could be rational. Figures 3 and 4 show little support for that explanation.

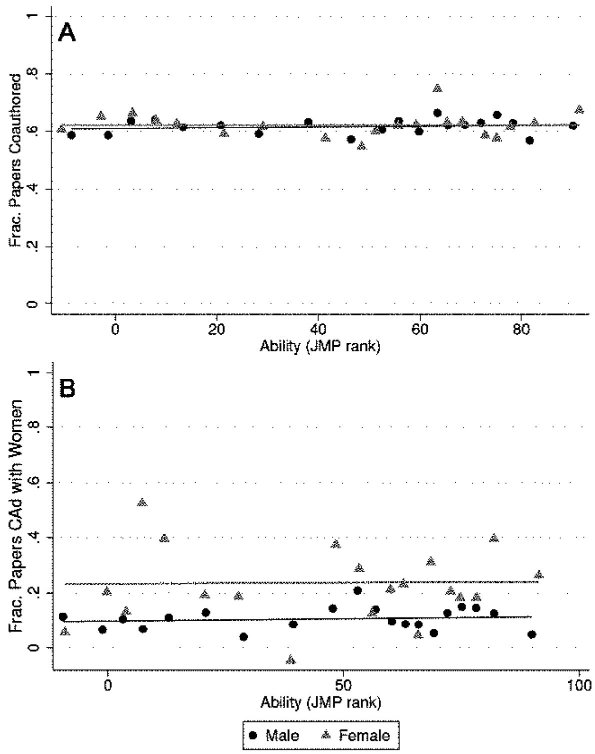


Fig. 3.—Ability and sorting. This binned scatterplot shows the correlation between an individual's ability and the propensity to coauthor (*A*) or the propensity to coauthor with women (*B*). The outcome variable in *A* is the fraction of an individual's papers that were published by tenure that are coauthored. The outcome variable in *B* is the fraction of an individual's pretenure papers that are coauthored (CAd) with only women. We proxy for an individual's ability with the rank of the journal in which the individual's job market paper was published. The plot is constructed as described in figure 1, with the *y*-axis variable residualized on the following controls before plotting: total solo- and coauthored papers, the number of years it took to go up for tenure, log citations, and tenure school group, tenure year group, and field fixed effects. The lines of best fit using OLS are shown separately for men and women. The estimates for *A* are $\beta = -0.0001$ (SE = 0.0003) for women and $\beta = 0.0002$ (SE = 0.0002) for men. The estimates for *B* are $\beta = -0.00004$ (SE = 0.0008) for women and $\beta = 0.0002$ (SE = 0.0003) for men. JMP = job market paper. A color version of this figure is available online.

(a) Figure 3: Ability and sorting.

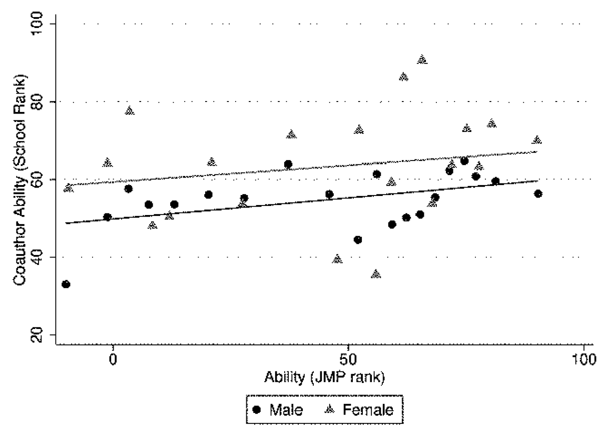


Fig. 4.—Assortative matching. This binned scatterplot shows the correlation between an individual's ability, proxied by the journal in which their job market paper is published, and their coauthor's ability, proxied by the average school rank of their coauthors. The school rank of coauthors is measured at the time that individual $i$ went up for tenure. School rankings are taken from RePEc/IDEAS. The plot is constructed as described in figure 1, with the *y*-axis variable residualized on the following controls before plotting: total solo- and coauthored papers, the number of years it took to go up for tenure, log citations, and tenure school group, tenure year group, and field fixed effects. The line of best fit using OLS is shown separately for men and women. The lines of best fit are estimated on the full sample and have slopes of $\beta = 0.062$ (SE = 0.091) for women and $\beta = 0.109$ (SE = 0.056) for men. JMP = job market paper. A color version of this figure is available

(b) Figure 4: Assortative matching.

## 6.7 Table 7 and Table 8: Accounting for sorting and top-five papers

**Read:**

- Table 7 adds controls for coauthor rank, rank differences, and the fraction of coauthors who are full professors.
- The female-by-coauthored interaction remains negative after accounting for these coauthor characteristics.
- Table 8 splits papers into top-five and non-top-five publications.
- Top-five coauthored papers help women more than non-top-five coauthored papers, suggesting that strong publication signals partly overcome uncertainty about contribution.

- Non-top-five coauthored papers do not have a positive estimated association with women's tenure probability.

**Economic message:** These tables show that the penalty is not merely about women working with more senior or higher-status coauthors. Very strong journal signals can partially discipline evaluator beliefs, but noisier coauthored signals remain problematic.

TABLE 7
ACCOUNTING FOR SORTING

| | Dependent Variable: Tenure | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| Solo authored | .086*** | .087*** | .084*** |
| | (.017) | (.017) | (.019) |
| Female × solo authored | .064*** | .069*** | .065*** |
| | (.017) | (.013) | (.015) |
| Coauthored | .078*** | .081*** | .074*** |
| | (.016) | (.014) | (.016) |
| Female × coauthored | −.037* | −.032* | −.034* |
| | (.016) | (.013) | (.014) |
| Female | −.139 | −.415*** | −.203 |
| | (.130) | (.112) | (.120) |
| Rank difference | .000 | | |
| | (.001) | | |
| Female × rank difference | −.000 | | |
| | (.001) | | |
| Average coauthor rank | | −.002 | |
| | | (.002) | |
| Female × average coauthor rank | | .005* | |
| | | (.002) | |
| Fraction full professor | | | −.044 |
| | | | (.067) |
| Female × fraction full professor | | | .044 |
| | | | (.066) |
| Observations | 540 | 567 | 567 |

NOTE.—The dependent variable in all columns is an indicator for receiving tenure. Column 1 shows the relationship between solo- and coauthored papers and tenure when controlling for the difference between individual *i*'s institution rank and the average institution rank of his or her coauthors. Column 2 controls for the average institution rank of an individual's coauthors, and col. 3 controls for the fraction of an individual's coauthors who are full professors. Only coauthors that an individual coauthored with up until tenure are included. All regressions control for tenure length, journal rank (AER-equivalent measure), and log citations. They also include tenure institution group, tenure year group, and field fixed effects. The sample size is smaller in this analysis because individuals with no coauthors are excluded.
* $p < .10$.
*** $p < .01$.

(a) Table 7: Accounting for sorting.

TABLE 8
PAPER SPLIT BY TOP FIVE

| | DEPENDENT VARIABLE: TENURE | |
|---|---|---|
| | Top Five (1) | Non–Top Five (2) |
| Solo | .070*** | .031*** |
| | (.017) | (.006) |
| Coauthored | .083*** | .029*** |
| | (.013) | (.006) |
| Female × solo | .017 | .060*** |
| | (.041) | (.016) |
| Female × coauthored | −.004 | −.039** |
| | (.039) | (.014) |
| Female | −.175* | |
| | (.088) | |
| Total coauthors | −.001 | |
| | (.005) | |
| Years to tenure | −.047*** | |
| | (.009) | |
| Log citations | .072*** | |
| | (.012) | |
| Tenure institution fixed effects | Yes | |
| Tenure year fixed effects | Yes | |
| Field fixed effects | Yes | |
| Observations | 612 | |
| $R^2$ | .397 | |

NOTE.—This table presents the results from estimating eq. (8). The results in the table are from this single regression, but solo- and coauthored papers are split into those published in the top five journals (col. 1) and journals below the top five (col. 2). Top five papers are those published in *AER*, *Econometrica*, *Journal of Political Economy*, *Quarterly Journal of Economics*, or *Review of Economic Studies*. The dependent variable is an indicator for receiving tenure. The regression includes tenure institution group, tenure year group, and field fixed effects. Robust standard errors are clustered by tenure institution and reported in parentheses.
* $p < .10$.
** $p < .05$.
*** $p < .01$.

(b) Table 8: Paper split by top five.

## 6.8 Table 9 and Table 10: Timing of coauthorship and Experiment I

**Read:**

- Table 9 tests whether women with slower early publication starts are more likely to begin coauthoring with men.
- The evidence does not support the claim that men bring struggling women onto projects near tenure.
- Table 10 reports Experiment I, in which predictors estimate future quiz scores after observing individual or joint score information.
- In math, when individual scores are visible, the female coefficient is not significant. When only joint scores are visible, women are predicted to score lower.
- In grammar, women are not penalized in the joint condition; when gender-performance information is shown, women receive higher predicted scores.

**Economic message:** Table 9 weakens a timing-based selection story. Table 10 supplies experimental evidence for the mechanism: when individual contributions are hidden, evaluators use gendered beliefs to assign credit.

TABLE 9
Timing of Coauthorship with Men

| | Years to First Publication (1) | Fraction of Papers with Men (2) |
|---|---|---|
| Female | −.050 | .137 |
| | (.172) | (.094) |
| Tenure | −.019 | .079 |
| | (.136) | (.046) |
| Female × tenure | −.037 | −.322* |
| | (.220) | (.140) |
| Years to first publication | | .015 |
| | | (.017) |
| Female × years to first publication | | −.063 |
| | | (.041) |
| Tenure × years to first publication | | −.032 |
| | | (.022) |
| Female × tenure × years to first publication | | .070 |
| | | (.064) |
| Total papers | −.114*** | −.006 |
| | (.016) | (.004) |
| AER equivalent | −.283 | .296** |
| | (.332) | (.091) |
| Tenure institution fixed effects | Yes | Yes |
| Tenure year fixed effects | Yes | Yes |
| Field fixed effects | Yes | Yes |
| Observations | 591 | 582 |

Note.—This table tests whether there are gender differences in the timing of an individual's first publication (col. 1) and whether women who take a longer time to publish their first paper are more likely to coauthor with men (col. 2). The outcome variable in col. 1 is the number of years it takes an individual to publish his or her first paper after graduating and is measured as the year of the individual's first publication minus the year of the individual's initial faculty appointment. Articles published before the first appointment (i.e., during graduate school) are not counted. The outcome variable in col. 2 is the fraction of an individual's papers published by tenure that are coauthored with men. The independent variable "Years to first publication" is the outcome variable in col. 1. Both regressions include tenure institution group, tenure year group, and field fixed effects. Robust standard errors are reported in parentheses.
* $p < .10$.
** $p < .05$.
*** $p < .01$.

(a) Table 9: Timing of coauthorship with men.

TABLE 10
Experiment I Predicted Score by Quiz Type

| | Dependent Variable: Predicted Quiz 2 Score | | | |
|---|---|---|---|---|
| | Math | | Grammar | |
| | Individual (1) | Joint (2) | Individual (3) | Joint (4) |
| Female | .111 | −.243** | −.021 | .103 |
| | (.081) | (.118) | (.071) | (.114) |
| Gender information | .145 | .076 | −.241** | −.456*** |
| | (.097) | (.134) | (.112) | (.130) |
| Female × gender information | −.108 | −.106 | .194 | .738*** |
| | (.115) | (.154) | (.131) | (.153) |
| Quiz 1 score | .735*** | .020 | .725*** | .016 |
| | (.057) | (.051) | (.066) | (.053) |
| Constant | .137 | 3.432*** | .289 | 3.246*** |
| | (.212) | (.361) | (.248) | (.386) |
| Observations | 250 | 266 | 231 | 262 |
| Predictors | 125 | 133 | 116 | 131 |
| $R^2$ | .298 | .041 | .239 | .139 |

Note.—This table presents the results from experiment I, in which participants predict how well an individual did on a math or grammar quiz on the basis of that individual's performance on an earlier quiz. Columns 1 and 2 show the results for the math quiz, and cols. 3 and 4 show the results from the grammar quiz. In the experiment, participants were randomized into the individual treatment, where participants saw each individual's score on a previous quiz (cols. 1 and 3), or the joint treatment, where participants saw the sum of two individuals' scores (cols. 2 and 4). "Gender information" is a dummy indicating that participants were told the average quiz scores of all men and women.
** $p < .05$.
*** $p < .01$.

(b) Table 10: Experiment I predicted score by quiz type.

### 6.9 Table 11: Experiment II

**Read:**

- Table 11 reports odds ratios from the HR recruiter experiment.
- Across all recruiters, candidate gender does not strongly affect selection because male and female recruiter biases partly offset each other.
- Male recruiters are less likely to choose female candidates in the joint-score setting, especially in the vocabulary task.
- Female recruiters are more likely to choose female candidates in the joint-score setting.
- Beliefs matter: recruiters who believe men are better at the task are less likely to choose female candidates, and recruiters who believe women are better are more likely to choose female candidates.

**Economic message:** The HR experiment shows that credit attribution bias is not unique to academic tenure committees. When output is joint and beliefs about ability differ, evaluators infer contribution through a gendered lens.

## 7 Short summary

- The paper studies whether gender affects recognition for group work.
- The main observational setting is tenure in economics, where solo-authored papers are clearer signals of individual contribution than coauthored papers.

TABLE 11
EXPERIMENT II ODDS RATIOS OF BEING PICKED BY TASK AND RECRUITER GENDER

| | Dependent Variable: All Recruiters | | Dependent Variable: Male Recruiters | | | | Dependent Variable: Female Recruiters | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Search | Vocabulary | Search | | Vocabulary | | Search | | Vocabulary | |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) |
| Female | 1.120 | 1.041 | .890 | .922 | .750** | .723** | 1.341** | 1.157 | 1.390** | 1.237 |
| | (.108) | (.099) | (.136) | (.142) | (.104) | (.098) | (.167) | (.152) | (.181) | (.178) |
| Female × belief | | | | .694*** | | .724*** | | .760*** | | .844* |
| | | | | (.072) | | (.072) | | (.081) | | (.086) |
| Joint score | 1.173*** | 1.032*** | 1.151*** | 1.149*** | 1.036*** | 1.036*** | 1.190*** | 1.189*** | 1.032*** | 1.032*** |
| | (.026) | (.003) | (.036) | (.037) | (.005) | (.005) | (.036) | (.037) | (.005) | (.005) |
| Candidate resumé controls | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Set × recruiter fixed effects | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Observations | 3,144 | 2,172 | 1,368 | 1,368 | 996 | 996 | 1,776 | 1,776 | 1,176 | 1,176 |
| Recruiters | 262 | 181 | 114 | 114 | 83 | 83 | 148 | 148 | 98 | 98 |

NOTE.—This table presents the results from experiment II, in which HR recruiters pick one candidate out of four for a search or vocabulary task on the basis of short résumés. Columns 1, 3, 4, 7, and 8 show the results for the search task, and cols. 2, 5, 6, 9, and 10 show the results for the vocabulary task. Results are shown separately depending on the recruiter's gender: all recruiters in cols. 1 and 2, male recruiters in cols. 3–6, and female recruiters in cols. 7–10. All regressions include fixed effects for each set-recruiter combination and controls for other variables in the candidates' résumés. Results are presented as odds ratios. Standard errors are clustered at the recruiter level.

* $p < .10$.
** $p < .05$.
*** $p < .01$.

Table 11: Experiment II odds ratios of being picked by task and recruiter gender.

- Women receive tenure at substantially lower rates than men, despite similar average numbers of total, solo-authored, and coauthored papers.
- Men benefit similarly from solo and coauthored work.
- Women benefit strongly from solo-authored work but less from coauthored work.
- The coauthor penalty for women is concentrated in papers coauthored with men.
- The estimated difference in credit assignment explains a large share of the observed gender tenure gap.
- Robustness checks show that the result is not driven by journal-rank measures, publication-window definitions, or sample attrition.
- Mechanism tests find little support for ability sorting, preference-based sorting, senior-coauthor composition, or timing stories.
- Experiment I shows that women receive less credit for joint performance on a male-stereotyped task when individual contributions are hidden.
- Experiment II shows that HR recruiters' gendered beliefs help explain candidate choices from joint outputs.
- The main lesson is that group production can create unequal promotion outcomes when evaluators cannot observe individual contributions and rely on stereotypes or priors.

# 8 Conclusion

## 8.1 Core conclusion

Sarsons, Gërxhani, Reuben, and Schram show that women receive less recognition for group work when individual contributions are difficult to observe. In the tenure data, women receive strong credit for solo-authored papers but less credit for coauthored papers, especially papers written with men. In the experiments, evaluators assign credit differently by gender when they see only joint output. The common mechanism is inference under uncertainty: when contribution signals are noisy, gendered beliefs affect attribution.

## 8.2 Economic intuition

The economic intuition is straightforward. A solo-authored paper is a relatively clean signal: the evaluator knows who produced it. A coauthored paper is a noisy signal: the evaluator must infer contribution shares. If evaluators hold priors that men are more likely to have made the key contribution in economics or in male-stereotyped tasks, women will receive less credit from joint work with men. Strong signals, such as solo-authored papers or top-five coauthored publications, can partially overcome these priors. Weaker or noisier signals leave more room for biased attribution.

## 8.3 Incremental contribution revisited

The paper advances the literature in three ways. First, it moves from documenting gender gaps to identifying a plausible micro-mechanism: unequal credit for team production. Second, it uses the institutional features of economics authorship to compare clear and noisy signals of contribution. Third, it combines observational tenure data with controlled experiments, allowing the authors to separate credit-attribution bias from selection into teams and actual effort differences.

## 8.4 Implications for evaluation and promotion

The findings imply that organizations should not treat team output as self-explanatory. Promotion committees, tenure committees, and hiring managers need explicit information about individual contributions. Contribution statements, transparent task records, letters that separately evaluate each collaborator's role, and standardized evaluation rubrics can reduce the room for biased inference. The point is not that group work is bad; the point is that group work requires better credit-allocation institutions.

## 8.5 Limitations

The tenure analysis is observational, so it cannot directly observe each coauthor's true contribution. The tenure variable is reconstructed from career movements rather than committee files. The experiments simplify real evaluation environments and use lower-stakes tasks. The academic setting may not generalize perfectly to all industries. Finally, the paper shows that credit-attribution bias exists and can be large, but it does not fully solve how organizations should optimally measure contributions in complex teams.

## 8.6 Takeaway

The enduring takeaway is that collaboration can be efficient but still inequitable if credit is allocated through biased inference. When individual contribution is visible, men and women are treated more similarly. When contribution is hidden inside group output, stereotypes and beliefs can determine who gets recognized. In settings where careers depend on recognition, that difference can become a promotion gap.